

MATH 20700 Honors Analysis I

Homework 4

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Problem 12

Proof. (a) (i) Let us write $D_n = \frac{f(\beta_n) - f(0) + f(0) - f(\alpha_n)}{\beta_n - \alpha_n} = \frac{f(\beta_n) - f(0)}{\beta_n - \alpha_n} + \frac{f(0) - f(\alpha_n)}{\beta_n - \alpha_n}$.

We can write $f'(0) = \lim_{n \rightarrow \infty} \frac{f(\beta_n) - f(0)}{\beta_n - 0} = \lim_{n \rightarrow \infty} \frac{f(\alpha_n) - f(0)}{\alpha_n - 0}$.

So let us define the following: $Q_\beta = \frac{f(\beta_n) - f(0)}{\beta_n - 0}$ and $Q_\alpha = \frac{f(\alpha_n) - f(0)}{\alpha_n - 0}$.

Then we can write $D_n = Q_\beta \cdot c_n + Q_\alpha \cdot d_n$, where $c_n = \frac{\beta_n}{\beta_n - \alpha_n}$ and $d_n = \frac{-\alpha_n}{\beta_n - \alpha_n}$. We note that $c_n + d_n = 1$.

If we write $Q_\beta = f'(0) + \epsilon_n$ and $Q_\alpha = f'(0) + \delta_n$ then we note that we must have $\lim_{n \rightarrow \infty} \epsilon_n = 0 = \lim_{n \rightarrow \infty} \delta_n$.

Let us write $D_n = c_n \cdot (f'(0) + \epsilon_n) + d_n \cdot (f'(0) + \delta_n) = (c_n + d_n) \cdot f'(0) + c_n \epsilon_n + d_n \delta_n$.

Then taking limits, we get

$$\lim_{n \rightarrow \infty} D_n = \lim_{n \rightarrow \infty} [(c_n + d_n)f'(0)] + \lim_{n \rightarrow \infty} c_n \epsilon_n + \lim_{n \rightarrow \infty} d_n \delta_n = f'(0).$$

In the last part, we know $\lim_{n \rightarrow \infty} c_n \epsilon_n = \lim_{n \rightarrow \infty} d_n \delta_n = 0$ because $c_n + d_n = 1$ hence both c_n, d_n are bounded, and we know that $\lim_{n \rightarrow \infty} \epsilon_n = 0 = \lim_{n \rightarrow \infty} \delta_n$.

- (ii) We use the same algebraic definitions as in part i), except we now observe that c_n is positive and d_n is negative. Further, $c_n + d_n = 1$ still holds, hence $0 < c_n \leq M$ tells us that $0 > d_n \geq 1 - M$.

Then, when we construct the same ϵ_n, δ_n as in part i), the same argument from part i) holds, because we still have the key result that c_n, d_n are bounded. Hence, we know

$$\lim_{n \rightarrow \infty} c_n \epsilon_n = \lim_{n \rightarrow \infty} d_n \delta_n = 0.$$

Therefore, $\lim_{n \rightarrow \infty} D_n = f'(0)$.

- (iii) By Mean Value Theorem, $\exists \theta \in (\alpha_n, \beta_n)$ such that $f'(\theta) = \frac{f(\beta_n) - f(\alpha_n)}{\beta_n - \alpha_n} = D_n$.

Taking limits, we observe that as $n \rightarrow \infty$, we have $\alpha_n \rightarrow 0$ and $\beta_n \rightarrow 0$. Since $\alpha_n < \theta < \beta_n$, we see by Squeeze Theorem that as $n \rightarrow \infty$, we have $\theta \rightarrow 0$.

Since the derivative f' exists and is continuous everywhere on $(-1, 1)$, we can pass limits inside the derivative. Hence, $\lim_{n \rightarrow \infty} D_n = f'(0)$.

- (b) If $f(x) = x^2 \sin \frac{1}{x}$, we see that $f'(x) = 2x \sin \frac{1}{x} - \cos \frac{1}{x}$.

We observe that $f'(x)$ is discontinuous at $x = 0$ because the $2x \sin \frac{1}{x}$ is bounded but the $\cos \frac{1}{x}$ term oscillates. Learning from part (a) (iii) above, we want to use this fact that the derivative is discontinuous at $x = 0$.

Let us try to find a sequence α_n such that $f(\alpha_n) = 0$ for all n . We observe that $\sin(2n\pi) = 0$ for all n , hence we construct the sequence $\alpha_n = \frac{1}{2n\pi}$. Clearly, $\lim_{n \rightarrow \infty} \alpha_n = 0$.

Let us now try to find a sequence β_n such that $f(\beta_n) = \beta_n^2$ for all n . We observe that $\sin(2n\pi + \frac{\pi}{2}) = 1$ for all n , hence we construct the sequence $\beta_n = \frac{1}{2n\pi + \frac{\pi}{2}}$. Clearly, $\lim_{n \rightarrow \infty} \beta_n = 0$.

By construction,

$$D_n = \frac{f(\beta_n) - f(\alpha_n)}{\beta_n - \alpha_n} = \frac{\beta_n^2}{\frac{-1}{2\pi(4n^2+n)}} = \frac{-2\pi(4n^2+n)}{(2n\pi + \frac{\pi}{2})^2} = \frac{-8\pi n^2 - 2\pi n}{4\pi^2 n^2 + 2\pi^2 n + \frac{\pi^2}{4}}.$$

Hence, taking the limit $n \rightarrow \infty$, we see that $\lim_{n \rightarrow \infty} D_n = -\frac{2}{\pi} \neq 0 = f'(0)$.

QED

Problem 17

Proof. For ease of notation, we will rewrite this as $\varphi(x) = \begin{cases} e^{-\frac{1}{x}} & \text{if } x > 0 \\ 0 & \text{if } x \leq 0. \end{cases}$

- (a) If $x < 0$, then $\varphi(x) = 0$ which is a constant function, hence $\varphi'(x) = 0$ hence φ' is also a constant function. By induction, if $\varphi^{(n)}$ is a constant function then $\varphi^{(n+1)}(x) = 0$. Hence, $\forall n, \forall x < 0, \varphi^{(n)}(x) = 0$.

We want to check that $\varphi'(0)$ exists, and for this we use the limit definition.

We have $\lim_{y \rightarrow 0^-} \frac{\varphi(y) - \varphi(0)}{y - 0} = \lim_{y \rightarrow 0^-} \frac{0}{y} = 0$.

For the other limit, we have $\lim_{y \rightarrow 0^+} \frac{\varphi(y) - \varphi(0)}{y - 0} = \lim_{y \rightarrow 0^+} \frac{e^{-\frac{1}{y}}}{y} = \lim_{y \rightarrow 0^+} \frac{1/y}{e^{1/y}} = \lim_{y \rightarrow 0^+} \frac{-1/y^2}{(-1/y^2) \cdot e^{1/y}} = \lim_{y \rightarrow 0^+} \frac{1}{e^{1/y}} = 0$ where we use L'Hopital's rule to evaluate the limit $\lim_{y \rightarrow 0^+} \frac{1/y}{e^{1/y}}$.

Now we want to show that φ is infinitely differentiable for $x > 0$ so we use induction. We claim that $\forall n, \forall x > 0, \varphi^{(n)}(x) = P_n(1/x) \cdot e^{-1/x}$ where $P_n(1/x)$ is some polynomial in $1/x$.

For the base case, $n = 1$, then $\varphi'(x) = x^{-2} \cdot e^{-1/x} = (1/x)^2 \cdot e^{-1/x}$.

For the inductive step, suppose the n -th derivative is of the form $\varphi^{(n)}(x) = P_n(1/x) \cdot e^{-1/x}$. Then

$$\begin{aligned} \varphi^{(n+1)}(x) &= \frac{d}{dx} [P_n(1/x) \cdot e^{-1/x}] \\ &= \left[\frac{d}{dx} P_n(1/x) \right] \cdot e^{-1/x} + P_n(1/x) \cdot \left[\frac{d}{dx} e^{-1/x} \right] \\ &= [P'_n(1/x) \cdot (-1/x^2)] \cdot e^{-1/x} + P_n(1/x) \cdot [(1/x^2) \cdot e^{-1/x}] \\ &= \left(-\frac{1}{x^2} P'_n(1/x) + \frac{1}{x^2} P_n(1/x) \right) \cdot e^{-1/x} \end{aligned}$$

We have hereby shown that φ is infinitely differentiable for $x < 0$ and $x > 0$. We have also shown that $\varphi'(0) = 0$. We now want to show that φ is infinitely differentiable at $x = 0$.

We want to show that if $\varphi^{(n)}(0) = 0$ then $\varphi^{(n+1)}(0) = 0$. We have already completed the base case.

For the inductive step, let us consider the limit definition of the derivative. $\varphi^{(n+1)}(0) = \lim_{y \rightarrow 0^-} \frac{\varphi^{(n)}(y) - \varphi^{(n)}(0)}{y - 0} = 0$.

Let us now check the other limit. $\varphi^{(n+1)}(0) = \lim_{y \rightarrow 0^+} \frac{\varphi^{(n)}(y) - \varphi^{(n)}(0)}{y - 0} = \lim_{y \rightarrow 0^+} \frac{\varphi^{(n)}(y)}{y} = \lim_{y \rightarrow 0^+} \frac{P_n(1/y) \cdot e^{-1/y}}{y}$.

Let $u = 1/y$, then we have $\lim_{y \rightarrow 0^+} \frac{P_n(1/y) \cdot e^{-1/y}}{y} = \lim_{u \rightarrow \infty} \frac{P_n(u) \cdot e^{-u}}{1/u} = \lim_{u \rightarrow \infty} \frac{u \cdot P_n(u)}{e^u} = \lim_{u \rightarrow \infty} \frac{Q(u)}{e^u} = \lim_{u \rightarrow \infty} \frac{C}{e^u} = 0$.

Here, in the last step, we apply L'Hopital's repeatedly (differentiating with respect to u) m times (where m is the degree of the polynomial $Q(u)$) to get a constant C in the numerator.

(b) No, φ is not analytic, because the Taylor polynomial around 0 is exactly 0 (because all derivatives are 0), but for $x > 0$, the Taylor polynomial is 0 whereas $\varphi(x) > 0$.

(c) $\beta(x) = e^2 \varphi(1-x) \cdot \varphi(x+1)$. Since φ is smooth for all x , we note that β is the product of two smooth functions (scaled up by the constant e^2) hence it is smooth.

Outside the interval: if $x \leq -1$ then $x+1 \leq 0$, hence $\varphi(x+1) = 0$ hence $\beta(x) = 0$. If $x \geq 1$ then $1-x \leq 0$ hence $\varphi(1-x) = 0$ hence $\beta(x) = 0$.

Inside the interval: if $-1 < x < 1$ then $0 < 1-x < 2$ hence $\varphi(1-x) > 0$ and $0 < x+1 < 2$ hence $\varphi(x+1) > 0$. Hence $\beta(x)$ is the product of positive numbers and the positive constant e^2 , hence it is positive.

$$\beta(0) = e^2 [\varphi(1)]^2 = e^2 (e^{-1})^2 = e^2 e^{-2} = e^0 = 1.$$

(d) For $-1 < x < 1$, we have $\beta(x) = e^2 \varphi(1-x) \varphi(x+1) = e^2 e^{-\frac{1}{1-x}} e^{-\frac{1}{1+x}}$.

$$\text{The exponent is } 2 - \left(\frac{1}{1-x} + \frac{1}{1+x}\right) = 2 - \frac{2}{1-x^2} = \frac{2x^2}{x^2-1}.$$

Hence, for $|x| < 1$, we have $\beta(x) = e^{2x^2/(x^2-1)}$.

QED

Problem 19

Proof. (a) Let us take a sequence of points (a_n) in D_k that converges to a^* (i.e., $a_n \rightarrow a^*$), then we want to show that $a^* \in D_k$.

Fix arbitrary $\delta > 0$, then $\exists n \in \mathbb{N}$ such that $a_n \in B_\delta(a^*)$. Since $B_\delta(a^*)$ is open, $\exists r > 0$ such that $B_r(a_n) \subset B_\delta(a^*)$.

Let $S_1 = \sup_{t \in B_r(a_n)} f(t)$ and $S_2 = \sup_{t \in B_\delta(a^*)} f(t)$. Then, we know that $S_1 \leq S_2$. Similarly, let $I_1 = \inf_{t \in B_r(a_n)} f(t)$ and $I_2 = \inf_{t \in B_\delta(a^*)} f(t)$. Then, we know that $I_1 \geq I_2$. Hence the oscillation $W(r) = S_1 - I_1$ in $B_r(a_n)$ is smaller than or equal to the oscillation $S_2 - I_2$ in $B_\delta(a^*)$.

If $r < r'$ then it is easy to see that $W(r) \leq W(r')$, since as r decreases, the sup in the ball of radius r cannot increase and the inf in the ball of radius r cannot decrease.

We know that $\lim_{r \rightarrow 0} W(r) = \text{osc}_{a_n} f \geq \frac{1}{k}$. Hence, for any fixed $r > 0$, we must have $W(r) \geq$

$$\lim_{r \rightarrow 0} W(r) = \text{osc}_{a_n} f \geq \frac{1}{k}.$$

Hence, $S_2 - I_2 \geq W(r) \geq \frac{1}{k}$. This completes the proof.

- (b) Let the discontinuity set of f be $\mathcal{D}_f$. Then, $\forall d \in \mathcal{D}_f$, we have $\text{osc}_d f > 0$, hence by the Archimedean Property, $\exists k \in \mathbb{N}$ such that $\text{osc}_d f > \frac{1}{k} > 0$. Then, $d \in D_k$. Thus, we can write $\mathcal{D}_f = \bigcup_{k \in \mathbb{N}} D_k$ which is a countable union of closed sets.

- (c) Let the set of continuity points be $\mathcal{C}_f = [a, b] \setminus \mathcal{D}_f$. Then, $\mathcal{C}_f = [a, b] \setminus \bigcup_{k \in \mathbb{N}} D_k = \bigcap_{k \in \mathbb{N}} ([a, b] \setminus D_k)$ which is a countable intersection of open sets.

QED

Problem 36

Proof. (a) Let J be the set of all jump discontinuities. Let us write $J = J_1 \cup J_2$ where $J_2 = \{j \in \mathbb{R} : |f(j^+) - f(j^-)| > 0\}$ and $J_1 = \{j \in \mathbb{R} : |f(j^+) - f(j^-)| = 0 \text{ and } |f(j^+) - f(j)| > 0\}$.

It suffices to show that both J_1 and J_2 are countable.

First let us consider $j \in J_1$. We will split it into two cases. For case 1, $f(j^+) > f(j^-)$, and for case 2, $f(j^+) < f(j^-)$. We will only prove case 1, because case 2 is analogous.

We know that $f(j^+) > f(j^-)$, so by density of rationals, we can find a rational number $q \in \mathbb{Q}$ such that $f(j^+) > q > f(j^-)$.

So let us now use this idea and flip it to construct the following. Fix $q \in \mathbb{Q}$, then consider the set $J_q \subset J_1$ defined as $J_q = \{j \in J_1 : f(j^-) < q < f(j^+)\}$. If $J_q = \emptyset$ then there is nothing to consider. Hence, consider $q \in \mathbb{Q}$ such that $J_q \neq \emptyset$, then consider $j_1 \in J_q$.

Then, there exists an interval $B_\delta^-(j_1) = (j_1 - \delta, j_1)$ such that $\forall x \in B_\delta^-(j_1), f(x) < q$. And similarly, there exists an interval $B_\delta^+(j_1) = (j_1, j_1 + \delta)$ such that $\forall x \in B_\delta^+(j_1), f(x) > q$.

Now the idea is that we consider this interval and pick a rational it, $p \in B_\delta(j_1)$. Then, for all $j \in J_q$, there exists a corresponding interval $B_\delta(j)$ and its corresponding rational $p \in \mathbb{Q}$. We would like to claim that this correspondence is an injection $J_q \rightarrow \mathbb{Q}$. To support this

claim, we need to show that if $j_1, j_2 \in J_q$ with $j_1 \neq j_2$, then their corresponding intervals are disjoint. (Otherwise, we could have $p \in B_\delta(j_1) \cap B_\delta(j_2)$ and then we cannot claim injectivity).

So let us prove that if $j_1, j_2 \in J_q$ with $j_1 \neq j_2$, then their corresponding intervals are disjoint. Suppose for sake of contradiction that $j_2 \in B_\delta(j_1)$. Then either $j_2 \in B_\delta^-(j_1)$ or $j_2 \in B_\delta^+(j_1)$. Without loss of generality assume $j_2 \in B_\delta^-(j_1)$. Then, by construction, $\forall x \in B_\delta^-(j_1), f(x) < q$, which contradicts the jump-discontinuity assumption that $q < f^+(j_2)$.

We have thus shown that the intervals are disjoint, so the correspondence is indeed injective. Thus, J_q is countable (because each element in J_q can be injectively mapped to an element in $\mathbb{Q}$, namely to the p we constructed). Then, $J_1 = \bigcup_{q \in \mathbb{Q}} J_q$ is a countable union of countable

sets, hence it is countable. (Note that the δ 's of the intervals are not uniform but depend on the $j \in J_1$. However for ease of notation, we have used δ everywhere).

Next we want to show that J_2 is countable. The argument is very similar to the one above for J_1 .

We split into cases. For case 1, $f(j) > f(j^+) = f(j^-)$, and for case 2, $f(j) < f(j^+) = f(j^-)$. Again, we will only prove case 1, because case 2 is analogous.

Similar to above, construct $J_q \subset J_2$ such that $J_q = \{j \in J_2 : f(j^-) = f(j^+) < q < f(j)\}$. If $J_q = \emptyset$ then there is nothing to consider. Hence, consider $q \in \mathbb{Q}$ such that $J_q \neq \emptyset$, then consider $j_1 \in J_q$.

Then there exists an interval $B_\delta(j_1)$ such that $\forall x \in B_\delta(j_1) \setminus \{j_1\}, f(x) < q < f(j_1)$. Pick a rational $p \in B_\delta(j_1)$. Again, it suffices to show that for $j_1, j_2 \in J_2$, their corresponding intervals are disjoint, $B_\delta(j_1) \cap B_\delta(j_2) = \emptyset$. (Again remember that the δ 's of the intervals are not uniform but depend on the $j \in J_2$. However for ease of notation, we have used δ everywhere).

Suppose for sake of contradiction that $j_2 \in B_\delta(j_1)$. Then, by construction, $\forall x \in B_\delta(j_1) \setminus \{j_1\}, f(x) < q$. However, $f(j_2) > q$ which is a contradiction. Hence, the intervals can be made disjoint.

By the same argument as above, we have made an injective map to $\mathbb{Q}$ so J_2 is a countable union of countable sets, hence it is countable.

- (b) This function has an oscillatory discontinuity at $x = 0$. This is because $\lim_{x \rightarrow 0^+} f(x)$ does not exist because the sin term oscillates.
- (c) The characteristic function of the rationals is discontinuous at every real number. To see this, pick $\epsilon < 1$ then no matter what $\delta > 0$ you choose, a ball of radius δ around a rational number will contain irrational numbers (and vice versa, a ball of radius δ around an irrational number will contain rational numbers), so the oscillation of the function in this ball of radius δ equals $1 > \epsilon$.

But since $\mathbb{R}$ is uncountable, and we have proved in part (a) that $f : \mathbb{R} \rightarrow \mathbb{R}$ has at most countably many jump discontinuities, we conclude that the discontinuities of the characteristic function of the rationals are of the second kind. (In this case, it is second-kind-discontinuous everywhere because the limit never exists in the first place - whereas jump discontinuity is defined in terms of limits *existing* but not equaling each other and/or the function's value).

QED

Problem 37

Proof. No. For a counterexample, consider the function $f = \begin{cases} 21 + \sin \frac{1}{x}, & \text{if } x > 0 \\ 0, & \text{if } x = 0 \\ \sin \frac{1}{x}, & \text{if } x < 0. \end{cases}$

Then, for $x > 0$, $20 \leq f(x) \leq 22$ and for $x < 0$, $-1 \leq f(x) \leq 1$ so the intermediate value 17 is never attained, i.e., $\forall x, f(x) \neq 17$. (And clearly the function is bounded).

However, the limits $\lim_{x \rightarrow 0^-} f(x)$ and $\lim_{x \rightarrow 0^+} f(x)$ do not exist, because $\sin \frac{1}{x}$ oscillates. Hence, it is not a jump discontinuity, because the limits do not exist in the first place. QED

Problem 57

Proof. Let $f(x) = \begin{cases} \frac{1}{x} + x, & \text{if } 1 \geq x > 0 \\ 0, & \text{if } x = 0 \\ \frac{1}{x}, & \text{if } -1 \leq x < 0. \end{cases}$

We see that the left side integral is

$$\int_{-1}^{-r} \frac{1}{x} dx = [\ln |x|]_{-1}^{-r} = \ln |-r| - \ln |-1| = \ln(r) - \ln(1) = \ln(r)$$

and the right side integral is

$$\int_r^1 \left(\frac{1}{x} + x\right) dx = [\ln |x| + \frac{1}{2}x^2]_r^1 = (\ln(1) + \frac{1}{2}) - (\ln(r) + \frac{1}{2}r^2) = \frac{1}{2} - \ln(r) - \frac{r^2}{2}.$$

Then, we see that

$$\begin{aligned} \lim_{r \rightarrow 0} \left(\int_{-1}^{-r} f(x) dx + \int_r^1 f(x) dx \right) &= \lim_{r \rightarrow 0} \left(\ln(r) + \frac{1}{2} - \ln(r) - \frac{r^2}{2} \right) \\ &= \lim_{r \rightarrow 0} \left(\frac{1}{2} - \frac{r^2}{2} \right) \\ &= \frac{1}{2}. \end{aligned}$$

However, we have

$$\int_{-1}^0 f(x) dx = \int_{-1}^0 \frac{1}{x} dx = \lim_{r \rightarrow 0^+} \int_{-1}^{-r} \frac{1}{x} dx = \lim_{r \rightarrow 0^+} \ln(r) = -\infty$$

hence the improper integral $\int_{-1}^1 f(x) dx$ does not exist.

Consider $g(x) = x$. Then, $\forall R$, $\int_{-R}^R g(x) dx = 0$ hence $\lim_{R \rightarrow \infty} \int_{-R}^R g(x) dx = 0$. However, the improper integral $\int_{-\infty}^{\infty} g(x) dx$ fails to exist because the integral $\int_0^{\infty} g(x) dx$ fails to exist.

QED

Problem 60

Proof. (a) Since $\sum a_n$ converges, we know that the sequence of partial sums $A_n = \sum_{i=1}^n a_i$ converges. We want to show that $\sum a_n b_n$ converges. It suffices to show that the sequence of partial sums $P_n = \sum_{i=1}^n a_i b_i$ is a Cauchy sequence.

Let us analyse the ‘tail’ terms

$$\begin{aligned}
\sum_{i=n+1}^m a_i b_i &= \sum_{i=n+1}^m (A_i - A_{i-1}) b_i \\
&= \sum_{i=n+1}^m (A_i b_i - A_{i-1} b_i) \\
&= (A_{n+1} b_{n+1} - A_n b_{n+1}) + (A_{n+2} b_{n+2} - A_{n+1} b_{n+2}) + \cdots + (A_m b_m - A_{m-1} b_m) \\
&= A_{n+1} (b_{n+1} - b_{n+2}) + A_{n+2} (b_{n+2} - b_{n+3}) + \cdots + A_{m-1} (b_{m-1} - b_m) + A_m b_m - A_n b_{n+1} \\
&= A_m b_m - A_n b_{n+1} + \sum_{i=n+1}^{m-1} A_i (b_i - b_{i+1}) \\
&= A_m b_m - A_n b_{n+1} + \sum_{i=n+1}^{m-1} (A_i - A_n) (b_i - b_{i+1}) + \sum_{i=n+1}^{m-1} A_n (b_i - b_{i+1}) \\
&= A_m b_m - A_n b_{n+1} + A_n (b_{n+1} - b_m) + \sum_{i=n+1}^{m-1} (A_i - A_n) (b_i - b_{i+1}) \\
&= A_m b_m - A_n b_m + \sum_{i=n+1}^{m-1} (A_i - A_n) (b_i - b_{i+1}) = (A_m - A_n) b_m + \sum_{i=n+1}^{m-1} (A_i - A_n) (b_i - b_{i+1}).
\end{aligned}$$

Now we want to show that $\forall \epsilon > 0, \exists N$ such that $\forall m > n > N$,

$$\left| \sum_{i=n+1}^m a_i b_i \right| = \left| (A_m - A_n) b_m + \sum_{i=n+1}^{m-1} (A_i - A_n) (b_i - b_{i+1}) \right| < \epsilon.$$

Let us apply Triangle Inequality to get

$$\left| \sum_{i=n+1}^m a_i b_i \right| \leq |(A_m - A_n) b_m| + \left| \sum_{i=n+1}^{m-1} (A_i - A_n) (b_i - b_{i+1}) \right|.$$

Since (A_n) is Cauchy, we can choose arbitrary $\epsilon^* > 0$ and replace $A_m - A_n$ and $A_i - A_n$ with ϵ^* to get

$$\begin{aligned} \left| \sum_{i=n+1}^m a_i b_i \right| &\leq |\epsilon^* b_m| + \sum_{i=n+1}^{m-1} |(A_i - A_n)| |b_i - b_{i+1}| \\ &\leq |\epsilon^* b_m| + \sum_{i=n+1}^{m-1} \epsilon^* |b_i - b_{i+1}| \\ &= |\epsilon^* b_m| + \epsilon^* |b_{n+1} - b_m|. \end{aligned}$$

Above, we use the fact that b_n is monotone to evaluate the telescoping sum expression. Since b_n is bounded, we know that $\forall k, |b_k| \leq M$. Hence, we can write

$$\left| \sum_{i=n+1}^m a_i b_i \right| \leq |\epsilon^* b_m| + \epsilon^* |b_{n+1} - b_m| \leq \epsilon^* M + \epsilon^* (2M) = 3M\epsilon^*.$$

Thus, picking $\epsilon^* < \frac{\epsilon}{3M}$ completes the proof.

(b) Let us pick $a_n = b_n = \frac{(-1)^n}{\sqrt{n}}$.

Then, by Alternating Series Test, $\sum a_n$ converges because $|a_{n+1}| < |a_n|$ and $a_n \rightarrow 0$.

Also, b_n is not monotone since the numerator $(-1)^n$ oscillates between 1 and -1 . Also $b_n \rightarrow 0$.

However, $a_n b_n = \frac{(-1)^{2n}}{(\sqrt{n})^2} = \frac{1}{n}$ and we know that $\sum a_n b_n = \sum \frac{1}{n}$ diverges.

QED

Problem 67

Proof. We remember that $\log(ab) = \log a + \log b$.

$$\log(C_n) = \log(c_1 c_2 \dots c_n) = \sum_{k=1}^n \log(c_k) = \sum_{k=1}^n \log(1 + a_k).$$

If $C_n \rightarrow C$ with $C > 0$, then $\log(C_n) \rightarrow \log(C)$ where $\log(C) > 0$ is a finite number.

Then, $\sum_{k=1}^n \log(1 + a_k) \rightarrow \log(C)$ hence it converges.

We have thus shown $\prod c_k$ converges $\iff \sum \log(1 + a_k)$ converges. Now we want to show that $\sum a_k$ converges $\iff \sum \log(1 + a_k)$ converges.

Let us try to use Limit Comparison Test to find the limit $L = \lim_{k \rightarrow \infty} \frac{\log(1 + a_k)}{a_k}$.

For the series $\sum a_k$ to converge, we must have $a_k \rightarrow_{k \rightarrow \infty} 0$ as a necessary (though not sufficient) condition.

Hence, to evaluate the limit L , let us consider the behaviour of the function $f(x) = \frac{\log(1+x)}{x}$ as $x \rightarrow 0$.

We can use L'Hopital's rule to get $L = \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = \lim_{x \rightarrow 0} \frac{1}{1+x} = 1$.

Thus, by Limit Comparison Test, since the limit L is a finite positive number, we know that $\sum a_k$ converges $\iff \sum \log(1+a_k)$ converges.

This completes the proof.

QED

Problem 72

Proof. (a) Let $S_N = \sum_{k=0}^N |c_k|$. The terms $|c_k|$ are all non-negative, so $\{S_N\}$ is a non-decreasing sequence of partial sums. By the Monotone Convergence Theorem, this sequence converges if and only if it is bounded above.

By definition, $c_k = a_0 b_k + a_1 b_{k-1} + \cdots + a_k b_0 = \sum_{i=0}^k a_i b_{k-i}$.

Applying the triangle inequality, we have

$$|c_k| = \left| \sum_{i=0}^k a_i b_{k-i} \right| \leq \sum_{i=0}^k |a_i| |b_{k-i}|.$$

Now we sum this inequality from $k = 0$ to N , so we get

$$S_N = \sum_{k=0}^N |c_k| \leq \sum_{k=0}^N \left(\sum_{i=0}^k |a_i| |b_{k-i}| \right)$$

For the double sum on the right, let $j = k - i$. The sum includes all terms $|a_i| |b_j|$ where $i \geq 0$, $j \geq 0$, and $i + j = k$ for k from 0 to N . This is the set of all pairs (i, j) such that $i + j \leq N$. Now, consider the product of the partial sums of the absolute series $A_N = \sum_{i=0}^N |a_i|$

and $B_N = \sum_{j=0}^N |b_j|$.

Their product is

$$A_N B_N = \left(\sum_{i=0}^N |a_i| \right) \left(\sum_{j=0}^N |b_j| \right) = \sum_{i=0}^N \sum_{j=0}^N |a_i| |b_j|.$$

This product sum includes all terms $|a_i| |b_j|$ where $0 \leq i \leq N$ and $0 \leq j \leq N$. If $i + j \leq N$ (the condition for the S_N sum), then it must be that $i \leq N$ and $j \leq N$ (since $i, j \geq 0$). Since all terms are non-negative, we have

$$\sum_{k=0}^N \left(\sum_{i=0}^k |a_i| |b_{k-i}| \right) \leq A_N B_N$$

Combining our inequalities, we have

$$S_N = \sum_{k=0}^N |c_k| \leq A_N B_N$$

Let $A = \sum_{i=0}^{\infty} |a_i|$ and $B = \sum_{j=0}^{\infty} |b_j|$. Since A_N and B_N are non-decreasing sequences (they are sums of non-negative terms), they are bounded above by their limits, i.e., $A_N \leq A$ and $B_N \leq B$ for all N .

We can now find a fixed upper bound for S_N , namely $S_N \leq A_N B_N \leq AB$. The sequence of partial sums $\{S_N\}$ is non-decreasing and is bounded above by the finite number AB .

Therefore, the sequence $\{S_N\}$ converges. This proves that $\sum_{k=0}^{\infty} |c_k|$ converges, and thus $\sum_{k=0}^{\infty} c_k$ converges absolutely.

- (b) We formulate some algebraic laws for such products (commutativity, distributivity, and so on) below.

Commutativity: $A \cdot B = B \cdot A$

$$\left(\sum_{i=0}^{\infty} a_i \right) \left(\sum_{j=0}^{\infty} b_j \right) = \left(\sum_{j=0}^{\infty} b_j \right) \left(\sum_{i=0}^{\infty} a_i \right)$$

Distributivity: $A \cdot (B + C) = (A \cdot B) + (A \cdot C)$

$$\left(\sum_{i=0}^{\infty} a_i \right) \left(\sum_{j=0}^{\infty} b_j + \sum_{k=0}^{\infty} c_k \right) = \left(\sum_{i=0}^{\infty} a_i \right) \left(\sum_{j=0}^{\infty} b_j \right) + \left(\sum_{i=0}^{\infty} a_i \right) \left(\sum_{k=0}^{\infty} c_k \right)$$

Associativity: $(A \cdot B) \cdot C = A \cdot (B \cdot C)$

$$\left(\left(\sum a_i \right) \left(\sum b_j \right) \right) \left(\sum c_k \right) = \left(\sum a_i \right) \left(\left(\sum b_j \right) \left(\sum c_k \right) \right)$$

Multiplicative Identity: There exists an identity series $I = \sum_{i=0}^{\infty} \delta_i$ such that $A \cdot I = A$. This identity series is $I = 1 + 0 + 0 + \dots$, where $\delta_0 = 1$ and $\delta_i = 0$ for $i > 0$.

Proof for Commutativity: We show that $[A \cdot B]_n = [B \cdot A]_n$ for all $n \geq 0$. By definition, the n -th term of $A \cdot B$ is:

$$[A \cdot B]_n = \sum_{i=0}^n a_i b_{n-i}$$

We perform a change of index by letting $j = n - i$. As i runs from 0 to n , j runs from n to 0.

$$\sum_{i=0}^n a_i b_{n-i} = \sum_{j=n}^0 a_{n-j} b_j = \sum_{j=0}^n b_j a_{n-j}$$

This final expression, $\sum_{j=0}^n b_j a_{n-j}$, is precisely the definition of the n -th term of the Cauchy product $B \cdot A$. Thus, $[A \cdot B]_n = [B \cdot A]_n$.

Proof for Distributivity: We show that $[A \cdot (B + C)]_n = [(A \cdot B) + (A \cdot C)]_n$ for all $n \geq 0$. First, the n -th term of the series $B + C$ is $(b_n + c_n)$. Now, we find the n -th term of the left-hand side, $A \cdot (B + C)$:

$$[A \cdot (B + C)]_n = \sum_{i=0}^n a_i \cdot [B + C]_{n-i} = \sum_{i=0}^n a_i (b_{n-i} + c_{n-i}) = \sum_{i=0}^n (a_i b_{n-i} + a_i c_{n-i})$$

where we used the distributive law for numbers at the last step. Since this is a finite sum, we can split it into two:

$$\left(\sum_{i=0}^n a_i b_{n-i} \right) + \left(\sum_{i=0}^n a_i c_{n-i} \right)$$

This is exactly $[A \cdot B]_n + [A \cdot C]_n$, which is the n -th term of the right-hand side series $(A \cdot B) + (A \cdot C)$. Thus, the n -th terms are equal.

QED

Problem 74

Proof. Each factor in the infinite product can be written as

$$\frac{1}{1 - p_k^{-s}} = \sum_{j=0}^{\infty} (p_k^{-s})^j = \sum_{j=0}^{\infty} \frac{1}{(p_k^j)^s} = 1 + \frac{1}{p_k^s} + \frac{1}{p_k^{2s}} + \frac{1}{p_k^{3s}} + \dots$$

Now, let's consider the N^{th} partial product, P_N , which includes all primes from p_1 up to p_N

$$P_N = \prod_{k=1}^N \frac{1}{1 - p_k^{-s}} = \prod_{k=1}^N \left(\sum_{j=0}^{\infty} \frac{1}{p_k^{js}} \right).$$

Since each of these N series converges absolutely, we are allowed to multiply them and rearrange the terms in any order as per Mertens's Theorem for Cauchy products, extended to multiple series. When we multiply these terms, a general term in the resulting sum is formed by picking one term from each of the N series, i.e.,

$$\frac{1}{p_1^{j_1 s}} \cdot \frac{1}{p_2^{j_2 s}} \cdot \dots \cdot \frac{1}{p_N^{j_N s}} = \frac{1}{(p_1^{j_1} p_2^{j_2} \dots p_N^{j_N})^s}$$

where each $j_k \geq 0$ is an integer.

The numbers we are constructing in the denominator, $n = p_1^{j_1} p_2^{j_2} \dots p_N^{j_N}$, are precisely all positive integers whose prime factors are only from the set $\{p_1, p_2, \dots, p_N\}$. Therefore, the partial

product P_N is the sum of $\frac{1}{n^s}$ for all such integers n , i.e., $P_N = \sum_{n \in S_N} \frac{1}{n^s}$ where $S_N = \{n \in \mathbb{Z}^+ \mid \text{all prime factors of } n \text{ are } \leq p_N\}$.

We want to show that $\lim_{N \rightarrow \infty} P_N = \zeta(s)$ so we consider the difference between $\zeta(s)$ and P_N .

$$\zeta(s) - P_N = \left(\sum_{n=1}^{\infty} \frac{1}{n^s} \right) - \left(\sum_{n \in S_N} \frac{1}{n^s} \right) = \sum_{n \notin S_N} \frac{1}{n^s}.$$

An integer n is not in S_N (i.e., $n \notin S_N$) if and only if it has at least one prime factor greater than p_N . The smallest such prime factor is p_{N+1} . This means any $n \notin S_N$ must be at least as large as p_{N+1} .

Therefore, the remaining sum is bounded by the tail of the full zeta series, i.e.,

$$0 < \zeta(s) - P_N = \sum_{n \notin S_N} \frac{1}{n^s} \leq \sum_{n=p_{N+1}}^{\infty} \frac{1}{n^s}$$

We know that the series $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$ converges for $s > 1$. A necessary condition for convergence is that the tail of the series must go to zero. As $N \rightarrow \infty$, $p_{N+1} \rightarrow \infty$ (since there are infinitely many primes). Thus, we have $\lim_{N \rightarrow \infty} \sum_{n=p_{N+1}}^{\infty} \frac{1}{n^s} = 0$. By the Squeeze Theorem,

since $0 < \zeta(s) - P_N \leq \sum_{n=p_{N+1}}^{\infty} \frac{1}{n^s}$, we must have $\lim_{N \rightarrow \infty} (\zeta(s) - P_N) = 0$. This implies that $\lim_{N \rightarrow \infty} P_N = \zeta(s)$. Hence, $\prod_{k=1}^{\infty} \frac{1}{1 - p_k^{-s}} = \zeta(s)$.

QED